

# ECHO

Emergent Collusion in Heterogeneous Oligopolies

## Progress Report - Practical Coding 60% Done / 40% Left

### 1. Project Information

|                   |                                                                                 |
|-------------------|---------------------------------------------------------------------------------|
| Student:          | Aryan Raj                                                                       |
| Project:          | ECHO - Algorithmic Collusion Simulation                                         |
| Repository:       | <a href="https://github.com/ARYANRAJ1121/ECHO">github.com/ARYANRAJ1121/ECHO</a> |
| Live Demo:        | <a href="https://echo-green-pi.vercel.app">echo-green-pi.vercel.app</a>         |
| Report Date:      | August 2026                                                                     |
| Overall Progress: | 60% practical implementation done                                               |

### 2. Status (coding only)

This report covers software implementation only. 60% = major practical coding finished and working in the repo. 40% = remaining coding / major features still to build. No viva, theory, or documentation tasks are counted here.

#### Progress bar



DONE 60% (core product)

LEFT 40% (more code)

| Block                                               | Share | Status       |
|-----------------------------------------------------|-------|--------------|
| Core engine + agents + regulator + UI + loaders     | 60%   | Done         |
| Live streaming, batch runs, GNN/XAI, richer markets | 40%   | Left to code |

### 3. Completed 60% (practical work done)

#### 3.1 Market engine

- MNL demand, Nash/monopoly solvers, Lambda index
- Bertrand loop + scale-invariant price bands

#### 3.2 Agents (5 architectures)

| Agent             | Module                          | Status             |
|-------------------|---------------------------------|--------------------|
| Heuristic control | <code>heuristic_agent.py</code> | Working ~0.15 L    |
| Q-Learning        | <code>rl_agent.py</code>        | Working            |
| DQN (NumPy NN)    | <code>dqn_agent.py</code>       | Working ~0.61-0.85 |
| LLM Groq Allam 2  | <code>llm_agent.py</code>       | Working            |
| RAG + pgvector    | <code>rag_agent.py</code>       | Working            |

#### 3.3 Detection pipeline

- Lambda monitor, NLP clustering, sentiment, demand shock
- Random Forest strategy classifier + price forecaster

### 3.4 Data loaders

| Dataset   | Source                | Type        |
|-----------|-----------------------|-------------|
| Gasoline  | BLS via FRED          | Live        |
| Crypto    | CoinGecko BTC history | Live        |
| Amazon    | Local listings CSV    | Real file   |
| Airlines  | DEL-BOM estimates     | Static code |
| Rideshare | Uber/Lyft estimates   | Static code |

### 3.5 Backend, DB, UI, automation

- FastAPI WebSocket server, PostgreSQL + pgvector, Docker Compose
- n8n webhooks, start\_echo.ps1, Chart.js dashboard + Vercel demo
- analysis plots / real\_data / CLI run\_simulation.py

### 3.6 Verified

- Gasoline / crypto / Amazon load with fallback=False
- End-to-end simulation + API import OK

## 4. Remaining 40% (coding / major work left)

All items below are software to implement - not report writing.

### A. Stronger live-data pipeline (~8%)

- Richer live feeds beyond current FRED + CoinGecko
- Stream live prices into dashboard beside the simulation
- Continuous live Lambda vs simulated Lambda + market alerts

### B. Experiment automation (~7%)

- JSON/YAML configs, seeded batch runner, result datasets
- Docker one-command full reproduce + CI tests
- Extra metrics: welfare loss, consumer surplus, Gini

### C. Advanced detection GNN / XAI (~8%)

- Graph Neural Network over firm interaction graph
- Temporal collusion graph, ringleader attention, autoencoders
- SHAP explainability + auto NL detection reports

### D. Richer market simulation (~10%)

- Multi-LLM backends in one market; capacity / entry-exit
- Multi-product bundling; agent communication channels
- Dynamic demand shocks; regulatory intervention module
- Multi-market simulation with cross-market spillover

### E. Product polish code (~7%)

- Wire n8n to Slack/email/Discord nodes
- Responsive/PWA dashboard + push alerts
- Deeper automated RAG long-run A/B runner

## 5. Run what is already built

cd antitrust\_sim

.\start\_echo.ps1 quick

Open <http://127.0.0.1:8000> -> Gasoline/Crypto/Amazon -> Heuristic or DQN

## 6. Sign-off

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**Practical major work done:** 60%

**Coding / major work left:** 40%

**Demoable today:** Yes

**Live/real data now:** Yes (gasoline, crypto, Amazon)

*Covers practical software implementation only (August 2026).*